

Module -1: Understanding of Hardware and Its Components

Section 1: Multiple Choice

1. Which of the following is NOT a component of the CPU?

1. ALU
2. RAM
3. CU
4. 1 and 3 both

ANS : 2. RAM

2. What is the function of RAM in a computer?

ANS : The **function of RAM is to store data and instructions temporarily** that the CPU needs while performing tasks.

3. Which of the following is a primary storage device?

1. HDD
2. SSD
3. SD card
4. 1 and 2 both

ANS : none of the above

4. What is the purpose of a GPU?

ANS : The **purpose of a GPU is to handle and accelerate graphics rendering and visual computations** in a computer system.

Section 2: True or False

5. True or False: The motherboard is the main circuit board of a computer where other components are attached.

ANS : TRUE

6. True or False: A UPS (Uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails.

ANS : TRUE

7. True or False: An expansion card is a circuit board that enhances the functionality of a component.

ANS : TRUE

Section 3: Short Answer

8. Explain the difference between HDD and SSD.

ANS : **HDD** : Uses **magnetic disks (platters)** to store data.

Has **moving mechanical parts** like a spinning disk and read/write head.

Slower in speed and more prone to damage due to movement.

Cheaper and available in **larger storage capacities**.

SSD : Uses **flash memory chips** to store data (no moving parts).

Faster in reading/writing data and booting the system.

More durable, energy-efficient, and **quieter** than HDDs.

Costlier per GB compared to HDDs.

9. Describe the function of BIOS in a computer system.

ANS : The **BIOS** is firmware stored on the motherboard that **initializes and tests hardware components** during computer startup.

- It **loads the operating system** from the storage device into memory.
- Functions of BIOS include:
- **Power-On Self-Test (POST)** – checks hardware on startup.
- **Bootstrap loader** – starts the OS.
- **Hardware configuration** – allows users to set up system settings.

10. List and briefly explain three input devices commonly used with Computers.

ANS : **Three commonly used input devices:**

1. **Keyboard:** Used to input text, numbers, and commands into the computer.
2. **Mouse:** A pointing device that controls the movement of the cursor and allows users to select, drag, and open items.
3. **Scanner:** Converts physical documents or images into **digital format** for editing or storage on a computer.

Section 4: Practical Application

11. Identify and label the following components on a diagram of a motherboard:

- CPU
- RAM slots
- SATA connectors¹².
- PCI-E slot

ANS : **CPU (Central Processing Unit) Socket:**

Usually located near the center or top of the motherboard.

It is a **square socket** where the **processor** is installed.

Surrounded by a **CPU fan or heat sink**.

- **RAM Slots (Memory Slots):**

Long and narrow slots usually placed **next to the CPU socket**.

Typically **two or four slots**, often color-coded (for dual-channel memory).

Used to install **DDR4 or DDR5 memory modules**.

- **SATA Connectors (Serial ATA Ports):**

Small **L-shaped connectors**, generally located near the **edge** of the motherboard.

Used to connect **storage devices** such as **HDDs and SSDs**.

- **PCI-E (Peripheral Component Interconnect Express) Slots:**

Long slots placed **below the CPU socket**.

Used to install **expansion cards**, such as **graphics cards (GPU)**, **network cards**, or **sound cards**.

12. Demonstrate how to install a RAM module into a computer.

ANS :

- **Power Off and Unplug:**
- **Open the Case.**
- **Locate RAM Slots.**
- **Open the Slot Clips.**
- **Align the RAM Module.**
- **Insert the RAM.**
- **Close the Case and Power On.**

Section 5: Essay

13. Discuss the importance of proper cooling mechanisms in a computer system. Include examples of cooling methods and their effectiveness.

ANS : Cooling is one of the most essential aspects of a computer's performance and reliability. When components like the **CPU, GPU, and power supply** operate, they generate heat. If this heat is not properly dissipated, it can lead to **overheating, reduced performance, and hardware damage**.

Importance:

1. **Prevents Overheating:**
 - a. Keeps temperature within safe limits to ensure stable performance.
2. **Improves Performance:**
 - a. Components like CPUs automatically **throttle speed** when overheated; good cooling prevents this slowdown.
3. **Increases Lifespan:**
 - a. Proper cooling reduces wear on sensitive electronic components.
4. **Enhances System Stability:**
 - a. Maintains consistent operation even during heavy workloads or gaming.

Examples of Cooling Methods:

1. **Air Cooling:**
 - a. Uses **fans and heat sinks** to transfer heat away from components.
 - b. Common and cost-effective.
 - c. Example: CPU fan and case exhaust fan.
2. **Liquid Cooling (Water Cooling):**
 - a. Uses a **liquid coolant** circulated through tubes and a radiator to absorb heat.
 - b. More **efficient** for high-performance systems but **expensive**.
 - c. Example: AIO (All-In-One) liquid coolers.
3. **Thermal Paste:**
 - a. Applied between the **CPU and heat sink** to improve thermal conductivity.
4. **Ventilation and Airflow Design:**
 - a. Proper **airflow inside the case** ensures cool air enters and hot air exits efficiently.

Conclusion:

Efficient cooling ensures **optimal performance, reliability, and durability** of computer components, making it a vital part of computer hardware design and maintenance.

14. .Explain the concept of bus width and its significance in computer Architecture.

ANS : A **bus** in computer architecture is a communication pathway that **transfers data, addresses, and control signals** between different components (like CPU, memory, and peripherals).

Bus Width:

- **Bus width** refers to the **number of bits** that can be transmitted simultaneously over the bus.
- For example, a **32-bit bus** can transfer 32 bits of data at once, while a **64-bit bus** can transfer 64 bits at once.

Significance:

1. Data Transfer Speed:

- a. A wider bus allows **more data** to move at once, improving system speed and efficiency.
- b. Example: A 64-bit processor can handle **double the data** of a 32-bit processor in a single cycle.

2. System Performance:

- a. Wider bus width enables **faster communication** between CPU, memory, and input/output devices.

3. Compatibility and Capacity:

- a. Determines how much **memory** a system can address.
- b. Example: 32-bit systems can typically address up to 4 GB RAM, while 64-bit systems can access **much larger memory** (theoretically up to 16 exabytes).

Conclusion:

Bus width plays a crucial role in defining the **speed, capacity, and performance** of a computer system. As technology evolves, wider bus architectures enable **faster data processing** and **enhanced multitasking capabilities**.